

Portfolio Management

The Behavioral Biases of Individuals

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LEARNING OUTCOMES

Mastery	The candidate should be able to:
☐	compare and contrast cognitive errors and emotional biases
☐	discuss commonly recognized behavioral biases and their implications for financial decision making
☐	describe how behavioral biases of investors can lead to market characteristics that may not be explained by traditional finance

INTRODUCTION

1

Research has demonstrated that when people face complex decisions, they often rely on basic judgments and preferences to simplify the situation rather than acting completely rationally. Although such approaches are quick and intuitively appealing, they may lead to suboptimal outcomes. In contrast to this body of research, traditional economic and financial theory generally assumes that individuals act rationally by considering all available information in the decision-making process, leading them to optimal outcomes and supporting the efficiency of markets. Behavioral finance challenges these assumptions by incorporating research on how individuals and markets actually behave. In this reading, we explore a foundational concept of behavioral finance: behavioral biases. Investment professionals may be able to improve economic outcomes by understanding these biases, recognizing them in themselves and others, and learning strategies to mitigate them.

The reading proceeds as follows. Section 2 describes and broadly characterizes behavioral biases. Sections 3 and 4 discuss specific behavioral biases within two broad categories: cognitive errors and emotional biases. The discussion includes a description of each bias, potential consequences, and guidance on detecting and mitigating the effects of the bias. Section 5 discusses market anomalies, which are essentially aggregate expressions of individual biases among financial market participants. A summary and practice problems conclude the reading.

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BEHAVIORAL BIAS CATEGORIES



compare and contrast cognitive errors and emotional biases

In general, behavioral biases come in two forms: faulty cognitive reasoning, known as **cognitive errors**, and those based on feelings or emotions, known as **emotional biases**. Both forms of bias, regardless of their source, may cause decisions to deviate from what is assumed by traditional finance theory.

Cognitive errors can often be corrected or eliminated through better information, education, and advice. Emotional biases, on the other hand, are harder to correct because they stem from impulses and intuitions. They arise spontaneously rather than through conscious effort and may even be undesired to the individual feeling them. Thus, it is often possible only to recognize an emotional bias and adapt to it. The cognitive–emotional distinction will help us determine when and how to adjust for behavioral biases in financial decision making.

Researchers have identified numerous behavioral biases. This reading does not attempt to catalog all of them. Rather, it discusses some of the more publicized and recognized biases within the cognitive–emotional framework. Additionally, we limit our focus to gauging the presence or absence—not the magnitude—of each bias discussed. That is, we will not try to measure how strongly the bias is exhibited, but rather we will describe the behavioral bias, its potential consequences, and the detection of and correction for the behavioral bias. In detecting a bias, we will identify statements or thought processes that may indicate the bias. Diagnostic tests of varying degrees of complexity are available to detect biases but are beyond the scope of this reading.

Finally, the individuals of interest in this reading are “financial market participants” (FMPs) engaged in financial decision making. These include both individual investors and financial services professionals.

3

COGNITIVE ERRORS



discuss commonly recognized behavioral biases and their implications for financial decision making

We classify cognitive errors into two categories: “belief perseverance biases” and “processing errors.”

Belief perseverance is the tendency to cling to one’s previously held beliefs by committing statistical, information-processing, or memory errors. The belief perseverance biases discussed are conservatism, confirmation, representativeness, illusion of control, and hindsight.

Processing errors describe how information may be processed and used illogically or irrationally in financial decision making. The processing errors discussed are anchoring and adjustment, mental accounting, framing, and availability.

Belief Perseverance Biases

Belief perseverance biases result from the mental discomfort that occurs when new information conflicts with previously held beliefs or cognitions, known as **cognitive dissonance**. To resolve this discomfort, people may ignore or modify conflicting information and consider only information that confirms their existing beliefs or thoughts.

Conservatism Bias

Conservatism bias is a belief perseverance bias in which people maintain their prior views or forecasts by inadequately incorporating new, conflicting information. In Bayesian terms, they tend to overweight their prior probability of the event and underweight the new information, resulting in revised beliefs about probabilities and outcomes that underreact to the new information.

Consequences of Conservatism Bias

As a result of conservatism bias, FMPs may do the following:

- Maintain or be slow to update a view or a forecast, even when presented with new information; and
- Maintain a prior belief rather than deal with the mental stress of updating beliefs given complex data. This behavior relates to an underlying difficulty in processing new information.

Detection of and Guidance for Overcoming Conservatism Bias

The effect of conservatism bias may be corrected for or reduced by properly analyzing and weighting new information. The first step is to be aware that a bias exists, especially about information that is technical, abstract, and/or statistical, because the **cognitive cost** involved in processing those forms of information is higher than for other types.

When new information is presented, the FMP should ask such questions as, “How does this information change my forecast?” or “What effect does this information have on my forecast?” FMPs should conduct careful analysis incorporating the new information and then respond appropriately. This updating of prior beliefs in light of new information is consistent with the tenets of Bayes’ Rule, in which updated probabilities are derived by systematically combining previous estimates and new information.

If information is difficult to interpret or understand, FMPs should seek guidance from someone who can either explain how to interpret the information or can explain its implications.

Confirmation Bias

Confirmation bias refers to the tendency to look for and notice what confirms prior beliefs and to ignore or undervalue whatever contradicts them. A response to cognitive dissonance, confirmation bias reflects a predisposition to justify to ourselves what we want to believe.

Most experienced private wealth advisers have dealt with a client who conducts some research and insists on adding a particular investment to the portfolio. The client may insist on continuing to hold the investment, even when the adviser recommends otherwise, because the client’s follow-up research seeks only information that confirms his belief that the investment is still a good value. The confirmation bias is not limited to individual investors; all FMPs should be wary of the potential confirmation biases within themselves.

EXAMPLE 1**Confirmation Bias**

A portfolio manager at Sarter Investment Advisors recommended shares of Real Media Inc., a hypothetical television production and distribution company, largely on the basis of compelling analytical and valuation work from a top equity research analyst. Sarter's clients have owned the shares for several years.

Recently, the shares have underperformed significantly as a result of the company missing analysts' earnings estimates and also in response to executive management turnover. The portfolio manager's colleagues believe this underperformance is a result of Real Media losing market share to a competitor with superior technology and distribution. The competitor is publicly traded, but Sarter's portfolio managers and analysts have not done research on it.

After another poor earnings release from Real Media, the portfolio manager speaks with the equity research analyst whose work was the primary source of the investment. The equity research analyst, who maintains a buy rating on the stock, believes that Real Media is now a more compelling investment than ever because its share price has fallen while its earnings estimates remain unchanged. As a result of the conversation, the portfolio manager feels reassured and holds the position.

The portfolio manager is subject to confirmation bias. Rather than speaking with a research analyst who has a sell rating on the stock or conducting research on the competitor to consider a different perspective, the portfolio manager speaks with someone who has an opinion she already shares.

Consequences of Confirmation Bias

As a result of confirmation bias, FMPs may do the following:

- Consider only the positive information about an existing investment while ignoring any negative information about the investment.
- Develop screening criteria while ignoring information that either refutes the validity of the criteria or supports other criteria. As a result, some good investments that do not meet the screening criteria may be ignored, and conversely, some bad investments that do meet the screening criteria may be made.
- Under-diversify portfolios. FMPs may become convinced of the value of a single company's stock. They ignore negative news, and they gather and process only information confirming that the company is a good investment. They build a larger position than appropriate and hold an under-diversified portfolio.
- Hold a disproportionate amount of their investment assets in their employing company's stock, because they believe in their company and are convinced of its favorable prospects. Favorable information is cited, and unfavorable information is ignored. If the employee were to acknowledge unfavorable information, the associated mental discomfort might make work very difficult for the employee.

Detection of and Guidance for Overcoming Confirmation Bias

The effect of confirmation bias may be corrected for or reduced by actively seeking out information that challenges existing beliefs.

Another useful step is to corroborate an investment decision. For example, if investment selections are based on criteria confirming an existing belief, such as stocks breaking through 52-week highs, it is usually advisable to corroborate that decision with research from another perspective or source (e.g., fundamental research on the company, industry, or sector).

Representativeness Bias

Representativeness bias refers to the tendency to classify new information based on past experiences and classifications. New information may resemble or seem *representative* of familiar elements already classified, but in reality, it can be very different. In these instances, the classification reflex can deceive, producing an incorrect understanding that biases all future thinking about the information. *Base-rate neglect* and *sample-size neglect* are two types of representativeness bias that apply to FMPs.

In **base-rate neglect**, a phenomenon's rate of incidence in a larger population—its base rate—is neglected in favor of specific information. The specific, individual information may be misleading relative to the more appropriate base rate or general information. FMPs often follow this erroneous path because diligent research is often conducted on an individual security or strategy, leading FMPs to overlook or ignore general information about the “class” to which an investment belongs, such as an industry, sector, or geography.

A second type of representativeness bias is **sample-size neglect**, in which FMPs incorrectly assume that small sample sizes are *representative* of populations. Individuals prone to sample-size neglect are quick to treat properties reflected in small samples as properties that accurately describe large pools of data, overweighting the information in the small sample.

EXAMPLE 2

Representativeness Bias

Jacques Verte is evaluating the future prospects of APM Company, a large auto parts manufacturer having some difficulties. During the last 50 years, very few auto part manufacturers have failed, even during periods of difficulty. A number of recent headlines have highlighted APM's business and financial difficulties, however, with some commentators suggesting that APM may go out of business.

1. Which of the following scenarios is more likely? Explain why.

- A. APM will solve its difficulties.
- B. APM will go out of business.

Solution:

Scenario A is more likely. The base rate, based on 50 years of data, is that more auto parts companies survive difficult times than fail. Thus, it is more likely that APM will solve its difficulties than go out of business.

2. If Verte is subject to representativeness bias, is he more likely to classify APM into A or B? Explain why.

Solution:

If Verte is subject to representativeness bias, he is likely to choose Scenario B, predicting that the company will go out of business because of the headlines he has read. In classifying APM as likely to go out of business, Verte

would be guilty of base-rate neglect by ignoring the low base rate of auto parts manufacturers failing even during times of difficulty.

Consequences of Representativeness Bias

As a result of representativeness bias, FMPs may do the following:

- Adopt a view or a forecast based almost exclusively on individual, specific information or a small sample; and
- Update beliefs using simple classifications rather than deal with the mental stress of updating beliefs given the high cognitive costs of complex data.

Detection and Guidance on Overcoming Representativeness Bias

When FMPs sense that base-rate or sample-size neglect may be a problem, they should ask the following question: “What is the probability that X (the investment under consideration) belongs to Group A (the group it resembles or is considered representative of) versus Group B (the group it is statistically more likely to belong to)?” This question, or a similar question, will help FMPs think through whether they are failing to consider base-rate probabilities or neglecting the law of small numbers and thus inaccurately assessing a particular situation. It may be necessary to do more research to obtain base-rate information and/or widen the sample size of observations.

Illusion of Control Bias

In **illusion of control bias**, people tend to believe that they can control or influence outcomes when, in fact, they cannot. Many researchers have uncovered situations where people perceived themselves as possessing more control than they did, inferred causal connections where none existed, or displayed surprisingly great certainty in their predictions for the outcomes of chance events. A classic example is that people prefer choosing their own lottery numbers over random numbers selected for them.

EXAMPLE 3

Illusion of Control Bias

Adelia Scott is a wealth adviser at Sarter Investment Advisors (Sarter), an investment advisory firm for high-net-worth individuals. Scott meets with a client who has 30% of his account in shares of his employer’s stock. The client is not subject to any employee holding requirement.

Prior meeting notes indicate that the client initially agreed to diversify the concentrated position over a five-year period. Scott recommends a faster schedule, however, based on recent research indicating that the company’s future growth prospects have considerably worsened as a result of industry trends and macroeconomic conditions.

When presented with this information, the client is reluctant to change his diversification plan, citing the company’s history of double-digit growth and his belief that this rate of growth will continue for the foreseeable future. The client remarks, “Trust me, my team and I are not going to let those forecasts you’re citing come true.”

The client is subject to illusion of control bias. He is unwilling to believe Scott’s opinion because he believes that he and his team can control the company’s performance and stock price.

Consequences of Illusion of Control

As a result of illusion of control bias, FMPs may do the following:

- Inadequately diversify portfolios. Research has found that some investors prefer to invest in companies that they feel they have control over, such as the companies they work for, leading them to hold concentrated positions. In fact, most investors have little or no control over the companies they work for. If the company performs poorly, these investors may experience both the loss of employment and investment losses.
- Trade more than is prudent. Researchers have found that portfolio turnover is negatively correlated with investment returns.
- Construct financial models and forecasts that are overly detailed. FMPs may require detailed models before making an investment decision, believing that forecasts from these models control uncertainty. Although a greater understanding of an investment, issuer, or industry is often useful, increased model complexity does not control the inherent risk and uncertainty of investment outcomes.

Detection of and Guidelines for Overcoming Illusion of Control Bias

The first and most basic idea that investors need to recognize is that investing is a probabilistic activity. Even the largest investment management firms have little control over the outcomes of the investments they make. Companies are subject to macroeconomic and industry forces, as well as the actions of competitors, customers, and suppliers.

Second, it is advisable to seek contrary viewpoints. As you contemplate a new investment, take a moment to think about considerations that might weigh against making the investment. Ask yourself: What are the downside risks? What might go wrong? When will I sell? It is often useful to speak with someone who has an opposing view, such as an equity research analyst with a sell rating on the subject stock.

Hindsight Bias

Hindsight bias refers to believing past events as having been predictable and reasonable to expect. This behavior results from the obvious fact that outcomes that did occur are more readily evident than outcomes that did not. Similarly, people tend to remember their own predictions of the future as more accurate than they actually were because they are biased by the knowledge of what actually occurred. Poorly reasoned decisions with positive results may be remembered as brilliant tactical moves, and poor results of well-reasoned decisions may be described as avoidable mistakes.

EXAMPLE 4

Hindsight Bias

Beverly Bolo, an analyst at an investment advisory firm, is giving a presentation to clients that, among other topics, explains the firm's investment results during past macroeconomic downturns. In the presentation, Bolo points out that the "occurrence of the last recession was obvious upon inspection of the yield curve and other leading indicators eight months before the downturn started."

Bolo's comment exhibits hindsight bias. Recessions, like any other event, appear obvious in hindsight but are hardly ever accurately predicted. Bolo could augment her remarks by exploring how often these leading indicators suggested that a recession is imminent against how often a recession subsequently occurred.

Consequences of Hindsight Bias

As a result of hindsight bias, FMPs may do the following:

- Overestimate the degree to which they correctly predicted an investment outcome, or the predictability of an outcome generally. This bias is closely related to overconfidence bias, which is discussed later in the reading.
- Unfairly assess money manager or security performance. Based on the ability to look back at what has taken place in securities markets, performance is compared against what has happened as opposed to expectations at the time the investment was made.

Detection of and Guidelines for Overcoming Hindsight Bias

Once understood, hindsight bias should be recognizable. FMPs should ask such questions as, “Am I re-writing history or being honest with myself about the mistakes I made?”

To guard against hindsight bias, FMPs should carefully record their investment decisions and key reasons for making those decisions in writing at or around the time the decision is made. Consulting these written records rather than memory will often produce a far more accurate examination of past decisions.

Processing Errors

Processing errors refer to information being processed and used illogically or irrationally. As opposed to belief perseverance biases, processing errors are less related to errors of memory or in assigning and updating probabilities; rather, they relate more closely to flaws in how information itself is processed.

Anchoring and Adjustment Bias

Anchoring and adjustment bias refers to relying on an initial piece of information to make subsequent estimates, judgments, and decisions. When required to estimate a value with unknown magnitude, people generally begin with some initial default number—an “anchor”—which they then adjust up or down. Regardless of how the initial anchor was chosen, people tend to adjust their anchors insufficiently and produce end approximations that are, consequently, biased. Anchoring and adjustment bias is closely related to conservatism bias. Bayes Rule again provides guidance for how new information should be incorporated into changing prior beliefs.

EXAMPLE 5

Anchoring and Adjustment Bias

Aiden Smythe is an equity research analyst at a brokerage firm. Smythe covers Industrial Lift Plc, a company that manufactures construction machinery. The company’s business is sensitive to macroeconomic conditions, particularly non-residential construction activity. Last year, Industrial Lift reported £1.00 in EPS amid mostly strong non-residential construction activity levels. Smythe is updating his EPS estimate for this year. Non-residential construction activity has severely declined in the last two months, and some economists fear that a recession is likely. As a result, Smythe forecasts that EPS will fall 10% from the prior-year level, publishing a £0.90 estimate for the year.

Smythe’s estimate exhibits anchoring and adjustment. Smythe’s anchor is the prior year’s EPS of £1.00, despite the possibility of a material change in underlying conditions.

Consequences of Anchoring and Adjustment Bias

As a result of anchoring and adjustment bias, FMPs may stick too closely to their original estimates when learning new information. This mindset is not limited to downside adjustments; the same phenomenon occurs with upside adjustments as well.

Detection of and Guidelines for Overcoming Anchoring and Adjustment Bias

The primary action FMPs can take is consciously asking questions that may reveal an anchoring and adjustment bias. Examples of such questions include, “Am I holding onto this stock based on rational analysis, or am I trying to attain a price that I am anchored to, such as the purchase price or a high water mark?” and “Am I making this forecast based on previously observed quantities or based on future expected conditions?”

It is important to remember that a company’s revenues and earnings for a given period reflect conditions in that period. If the conditions in the future differ from the past, revenues and earnings will likely differ as well, sometimes radically. Similarly, security prices reflect investors’ perception of the future at a given point in time; a given investor’s cost basis, past market levels, and other conditions based in the past are often irrelevant. FMPs should look at the basis for any investment recommendation to see whether it is anchored to previous estimates or some “default” number.

Mental Accounting Bias

Mental accounting bias refers to mentally dividing money into “accounts” that influence decisions, even though money is **fungible** (Thaler 1980). Despite traditional finance theory assuming that FMPs consider their entire portfolio holistically in a risk–return context, Statman (1999, 2008) contends that the difficulty individuals have in addressing the interaction of different investments leads to mental accounting: Investors construct portfolios in a layered pyramid format, with each layer addressing a specific financial goal.

EXAMPLE 6

Mental Accounting Bias

Kendra Liu, an individual investor, owns shares in New Horizons Ltd., a pharmaceutical company. A drug in that company’s research and development pipeline unexpectedly succeeds in a clinical trial, resulting in the shares doubling overnight. Liu sells half of her position in New Horizons and uses the proceeds to purchase shares of Cutting Edge Ltd., a small-cap biotechnology company. Liu believes that an investment in Cutting Edge is a “high-risk, high-reward bet” that could result in a total loss, but she is comfortable making the investment because she is using the proceeds from the sale of New Horizons Ltd., money that she did not expect to have anyway.

Liu’s investment in Cutting Edge exhibits mental accounting bias. She has sorted the gains from New Horizons into a mental account based on its source, even though the money is fungible. Liu should invest the proceeds in a manner consistent with a holistic portfolio strategy.

Consequences of Mental Accounting Bias

As a result of mental accounting bias, FMPs may do the following:

- Neglect opportunities to reduce risk by combining assets with low correlations. Offsetting positions across various portfolio layers or mental accounts can lead to suboptimal aggregate performance.

- Irrationally distinguish between returns derived from income and those derived from capital appreciation. Although many investors feel the need to preserve principal, they focus on the idea of spending income that the principal generates. As a result, many FMPs chase income streams, unwittingly risking principal in the process.
- Irrationally bifurcate wealth or a portfolio into investment principal and investment returns. Some FMPs may believe that greater risk can be taken with returns (from either income or capital appreciation) than the principal initially contributed. A common euphemism for this scenario, from the casino world, is “playing with house money.”

Detection of and Guidelines for Overcoming Mental Accounting Bias

An effective way to detect and overcome mental accounting behavior is to recognize its drawbacks. The primary drawback is that correlations between investments are not considered, leading to unintentional risk taking. FMPs should go through the exercise of combining all of their assets onto one spreadsheet (without headings or account labels) to see the holistic asset allocation. This exercise often produces information that is surprising when seen as a whole, such as higher cash balances than expected. The logical next step would be to create a portfolio strategy taking all assets into consideration.

Framing Bias

Framing bias is an information-processing bias in which a person answers a question differently based on the way in which it is asked or framed. It is often possible to frame a given decision problem in more than one way.

For example, a situation may be presented within a *gain* context (one in four start-up companies succeed) or within a *loss* context (three out of four start-ups fail). Given the first frame, an FMP may adopt a positive outlook and make venture capital investments. Given the second frame, the FMP might not.

Narrow framing occurs when people evaluate information based on a *narrow frame* of reference—that is, losing sight of the big picture in favor of one or two specific points. For example, an investor might focus solely on a company’s executive management team, overlooking or even dismissing other important properties such as industry characteristics, fundamental performance, and valuation.

EXAMPLE 7

Effects of Framing Bias

Decision-making frames are quite prevalent in the context of investor behavior. Risk tolerance questionnaires can demonstrate how framing bias may occur in practice and how FMPs should be aware of its effects.

Suppose an investor is to take a risk tolerance questionnaire for the purpose of determining which “risk category” she is in. The risk category will determine asset allocations and the appropriate types of investments. The following information is provided to each questionnaire taker:

Over a 10-year period, Portfolio ABC has averaged an annual return of 10% with an annual standard deviation of 16%. Assuming a normal return distribution, in a given year there is a 67% probability that the return will fall within one standard deviation of the mean, a 95% probability that the return will fall within two standard deviations of the mean, and a 99.7% probability that the return will fall within three standard deviations of the mean. Thus, there is a